09-10 第1篇

阅读理解（说明文） Why Is the Sea So Salty?

① If you have ever been on a beach, you have probably tasted a tiny drop of seawater on your lips and made a face. The sea is **salty** — sometimes so salty that it stings small cuts on your feet. But rain is fresh, rivers are fresh, lakes and well water are fresh. So where does all the salt in the ocean come from? And why doesn't the sea get **sweeter** over time as fresh rain keeps falling into it? These are questions scientists once found much harder than they look.

② To understand the salty sea, we must first look at **salt itself**. What we call "salt" in everyday language is a white powder on the dinner table. In science, it is really a large family of **chemical substances**. By far the most common kind in seawater is **sodium chloride** — the same stuff we use to flavour noodles. About **3.5%** of seawater, by weight, is made of various salts, and sodium chloride alone makes up around **three-quarters** of that amount. One litre of seawater contains about **35 grams** of salt: roughly the weight of **three middle-sized coins**.

③ The main source of ocean salt is surprising: **rain and rivers**. Fresh rain may taste pure to us, but it is never pure. As rain falls through the air, it picks up small amounts of **carbon dioxide** and other gases, which make it a **very weak acid**. When this weak acid touches rocks on land, it slowly **dissolves tiny pieces of minerals**, including sodium, chloride, calcium and magnesium. These pieces are washed down the rivers and finally carried into the sea.

④ If fresh water keeps flowing into the sea, why doesn't the ocean slowly become **sweeter** instead of saltier? The answer lies in a quiet imbalance. When seawater **evaporates** into clouds under the sun, **only the water leaves**; the salt stays behind. Every day, huge amounts of seawater turn into clouds over the ocean, drift over the land, fall as fresh rain, and eventually flow back to the sea — this time carrying a new tiny load of minerals. The water completes the circle; the salt does **not**.

⑤ The second big source of ocean salt is deep below the sea itself. In the **dark corners of the deep ocean floor**, there are long cracks where hot, mineral-rich water bursts out from inside the earth. These hot **underwater springs**, sometimes called "**black smokers**", add iron, sulphur and other minerals directly to the seawater. They also help explain why seas in some regions are slightly more salty than seas in others.

⑥ Not every sea has the same saltiness. The **Dead Sea**, between Israel and Jordan, is famous for being **about ten times saltier** than most oceans — so salty that a person can

lie on its surface without sinking. The reason is simple geography: many rivers flow into the Dead Sea, but no rivers flow **out** of it. The only way water leaves is by evaporation, which, as we said, does not take salt with it. Over thousands of years, the salt has piled up to an extreme level.

⑦ In contrast, the **Baltic Sea** in northern Europe is **much less salty** than the world's average. Many big rivers pour fresh water into it, while the narrow passages to the open ocean limit how much salty water can flow in. Some parts of the Baltic are so fresh that fish that normally live in lakes can also live in its waters.

⑧ The salt in the sea matters to human life, too. It supports the **special plants and animals** that can only live in salty water, from small jellyfish to huge whales. It also helps drive the great **ocean currents**: salty water is a little heavier than fresh water, and this weight difference pushes cold salty water downwards in some places and warm fresher water upwards in others. Without this quiet underwater traffic, the global climate would be very different from the one we live in.

⑨ So next time a wave brushes past your ankles, think of the long journey behind that single salty drop. It has been dissolved from a far-away rock, carried by rivers, hidden in a cloud, warmed by a hot deep-sea spring, and perhaps pushed halfway around the planet by invisible currents — all before it reached your foot. The sea is not only big and blue. It is also a **slow, patient library of salt**, written by rain, rivers, and the earth's own deep fire.

注释： sodium chloride 氯化钠；dissolve 溶解；evaporate 蒸发；underwater spring 海底温泉；current 洋流

根据文章内容，选择最佳答案。

1. About what percentage of seawater, by weight, is made of various salts?

A. About 0.5%.
B. About 3.5%.
C. About 10%.
D. About 35%.

2. According to paragraph 3, how does rain water become slightly acidic?

A. By mixing with rubbish in the streets.
B. By picking up small amounts of carbon dioxide and other gases while falling through the air.
C. By passing through a special water factory.
D. By being boiled before falling.

3. Why doesn't the sea become sweeter over time, even though fresh rivers keep flowing into it?

A. Because rivers do not really flow into the sea.
B. Because evaporation removes water but leaves the salt behind.
C. Because sea animals drink the salt.
D. Because the sea is very cold.

4. Why is the Dead Sea about ten times saltier than most oceans, according to the passage?

A. Because scientists keep adding salt to it every year.
B. Because many rivers flow into the Dead Sea but no rivers flow out of it, and water leaves only by evaporation.
C. Because it is a very small lake with no rivers at all.
D. Because its water is colder than other seas.

5. According to paragraph 8, what is one important role of salt in the sea?

A. It makes the water colourful.
B. It supports special salt-water plants and animals, and helps drive the great ocean currents by making cold salty water heavier.
C. It stops all fish from living there.
D. It makes the sea warmer at night.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第②段 "About 3.5% of seawater, by weight, is made of various salts"
【解析】海水含盐约 3.5%。A、C、D 均错。故选 B。

2.
【答案】B
【题型】细节理解题
【定位】第③段 "As rain falls through the air, it picks up small amounts of carbon dioxide and other gases, which make it a very weak acid"
【解析】雨在下落时吸收 CO_2 等气体而呈弱酸性。A、C、D 均错。故选 B。

3.
【答案】B
【题型】细节理解题
【定位】第④段 "only the water leaves; the salt stays behind"
【解析】蒸发只带走水，盐留下来。A、C、D 均与事实不符。故选 B。

4.
【答案】B
【题型】细节理解题
【定位】第⑥段 "many rivers flow into the Dead Sea, but no rivers flow out of it. The only way water leaves is by evaporation"
【解析】有入无出 + 蒸发累积导致高盐。A、C、D 均错。故选 B。

5.
【答案】B
【题型】细节理解题
【定位】第⑧段 "It supports the special plants and animals ... helps drive the great ocean currents"
【解析】盐支撑海洋生物并推动洋流。A、C、D 均与原文不符。故选 B。